PAYTM SAFECLEAR

Humanizing the Ambiguous 60 Seconds of UPI Point-of-Sale Payments

Track 2: AI-Powered Financial Journeys — “Simpler, Faster, and More Human”

By Team Ok Paytm

WHAT'S THE PROBLEM?

"Paisa kat gaya par mila nahi":

- Merchant Blindness by Design: Soundboxes are hardcoded to stay silent on pending states. to stop fraud, leaving honest shopkeepers blind.
- The Customer Panic SMS: The customer receives a debit SMS from their bank while the counter remains silent, creating instant distrust.
- The "Lazy UX" Blanket Lie: Apps show generic hedging across all failures: "Payment Failed. If money was deducted, refund in 3–5 days" — even for pre-debit declines (wrong PIN, low balance) where money never left!
- The Accidental Double-Spend: Panicked customers scan a second QR code. Gateway RRN checks fail to stop this because each new scan generates a new transaction ID.

PROPOSED SOLUTION

1. Dynamic Status QR ("Scan-to-Know") & 4-Digit Token:
   • Instantly surfaces an HMAC-signed QR code and a high-contrast 4-digit token (Token #4821) on the customer screen for 1-second merchant POS verification.
2. Sarvam AI Failure Explainer & Voice Vocalizer:
   • High-confidence verdict badges: "ZERO PAISA KATA" vs. "PAISA KATA HAI — CLEARANCE IN PROGRESS".
   • Soundbox vocalizes fraud-safe holding cues: "Token 4821 verify ho raha hai. Saman na dein."
3. Intent-Level Duplicate Guard & 1-Tap Refund:
   • Evaluates repeat payment intent within 120s; equips merchant POS with a 1-tap instant reversal button backed by an NPCI UDIR dispute certificate.
4. Intelligent Fallback (UPI Lite):
   • Automatically recovers low-balance (Z9) or bank timeouts via 1-tap on-device UPI Lite (< ₹500) with zero core-banking server latency.

ARCHITECTURE & AI STACK

The Sarvam AI Engine:

- sarvam-105b (Vernacular Reasoning): Translates 40+ raw ISO/NPCI error codes (ZM, Z9, U16, U68) into plain Hindi, Hinglish, Tamil, and English.
- bulbul-v4-flash (<250ms TTS): Ultra-fast Soundbox speech synthesis tuned for Indian retail accents and rupee currency pronunciation.
- saaras-v4 (Speech-to-Text): Low-latency voice dispute intake for vernacular voice inquiries.
- sarvam-m (Intent Guard): Evaluates Customer VPA + Merchant ID + Timestamp (<120s) to classify duplicate panic payments at 98% confidence.

Core Engineering:

- FastAPI & WebSockets: Bi-directional real-time session engine syncing Customer App and Merchant POS simultaneously.
- HMAC-SHA256: Cryptographic integrity for counter status tokens.
- NPCI UDIR Docketing: Instant automated dispute pre-packaging.

INTERACTIVE COUNTER SCENARIOS

Scenario A: The False Alarm (Z9 Insufficient Balance)

- Before: Silent soundbox, customer panics about missing ₹350.
- SafeClear: Green Badge "Zero Paisa Kata" + 1-Tap UPI Lite checkout. Settles in 1.8 seconds.

Scenario B: Bank Server Timeout (U68)

- Before: Standoff over sweets; merchant refuses goods on pending status.
- SafeClear: Phone auto-flips to Dynamic QR [Token #4821]. Soundbox announces holding state. Merchant scans QR, switch confirms, turns green.

Scenario C: Duplicate Guard & 1-Tap Instant Refund

- Before: Customer pays ₹400 twice in 30s; waits 5 days for refund.
- SafeClear: sarvam-m catches duplicate intent. Soundbox warns merchant. POS surfaces "Issue 1-Tap Refund" with instant credit.

USP

1. Zero Ambiguity, Zero Hedging: Replaces the lazy "refund in 3-5 days" lie with definitive money movement truth ("Zero Paisa Kata").
2. Fraud-Safe Hardware Voice: Solves merchant blindness without creating fraud loopholes — Soundbox explicitly says "Saman na dein" until 100% switch settlement.
3. Intent-Level vs Technical Deduplication: Catches human panic payments across different UPI apps, where standard RRN dedup is completely blind.
4. Physical Counter Attribution: The 4-digit token allows instant matching in crowded queues without handing over phones.
5. 100% Sovereign AI: Built on Sarvam AI for native Indic language understanding, compliant with Indian regulatory and data sovereignty standards.

IMPACT & BENEFITS

Business Impact:

- 85% Drop in Tier-1 Dispute Tickets: Eliminates customer inquiries for pre-debit declines and duplicate debits.
- Zero Counter Standoffs: Cuts average in-flight dispute time at queues from 3 minutes to under 15 seconds.
- Strengthens Soundbox Retention: Reinforces Paytm Soundbox as an indispensable, intelligent counter device.

Phased Rollout:

- Phase 1 (Software): Dynamic Status QR, "Zero Paisa Kata" badge, and Sarvam failure explainer on Paytm Consumer & Business apps.
- Phase 2 (Hardware Fleet): Over-the-air firmware update to the Soundbox fleet integrating bulbul-v4-flash voice de-escalation cues.
- Phase 3 (Ecosystem Integration): Direct automated UDIR escalation and merchant instant-refund liquidity pools for enterprise retail.